d) **Paedogamy**- It is the sexual union of young individuals produced immediately after the division of the adult parent cell by mitosis.

e) **Merogamy**- The fusion of small sized and morphologically different gametes (merogametes) takes place.

f) **Isogamy**- The fusion of morphological and physiological identical gametes (isogametes) is called **isogamy**. e.g. *Monocystis.*

g) **Anisogamy**- The fusion of dissimilar gametes is called **anisogamy** (*Gr. An*-without; iso-equal; *gam*-marriage). Anisogamy occurs in higher animals but it is customary to use the term fertilization instead of anisogamy or syngamy. e.g. higher invertebrates and all vertebrates.

Conjugation is the temporary union of the two individuals of the same species. During their union both individuals, called the conjugants exchange certain amount of nuclear material (DNA) and then get separated. Conjugation is common among ciliates, e.g. *Paramecium, Vorticella* and bacteria (Prokaryotes).

Phases of life cycle: Organisms have three phases – Juvenile phase, reproductive phase and senescent phase. **Juvenile phase/ vegetative phase** is the period of growth between the birth of the individual upto reproductive maturity. During **reproductive phase/ maturity phase** the organisms reproduce and their offsprings reach maturity period. On the basis of time, breeding animals are of two types: **seasonal breeders** and **continuous breeders**. Seasonal breeders reproduce at particular period of the year such as frogs, lizards, most birds, deers etc., Continuous breeders continue to breed throughout their sexual maturity e.g. honey bees, poultry, rabbit etc., **Senescent phase** begins at the end of reproductive phase when degeneration sets in the structure and functioning of the body.

PARTHENOGENESIS (*Gr. Parthenos* – virgin, *Genesis*- produce)

Development of an egg into a complete individual without fertilization is known as parthenogenesis. It was first discovered by Charles Bonnet in 1745. Parthenogenesis is of two main types namely, Natural Parthenogenesis and Artificial Parthenogenesis. In certain animals, parthenogenesis occurs regularly, constantly and naturally in their life cycle and is known as **natural parthenogenesis**.



Natural parthenogenesis are of different types:

a) Arrhenotoky: In this type only males are produced by parthenogenesis. eg: honey bees

b) Thelytoky: In this type only females are produced by parthenogenesis.eg: Solenobia

c) Amphitoky: In this type parthenogenetic egg may develop into individuals of any sex. eg: *Aphis*

Natural parthenogenesis may be of two types, viz., complete and incomplete. **Complete parthenogenesis** is the only form of reproduction in certain animals and there is no biparental sexual reproduction. There are no male organisms and so, such individuals are represented by females only. **Incomplete parthenogenesis** is found in some animals in which both sexual reproduction and parthenogenesis occurs. e.g. In honeybees; fertilized eggs (zygotes) develop into queen and workers, whereas unfertilized eggs develop into drones (male). In **paedogenetic parthenogenesis (paedogenesis)** the larvae produce a new generation of larvae by parthenogenesis. It occurs in the sporocysts and Redia larvae of liver fluke. It is also seen in the larvae of some insects. e.g. Gall fly. In **artificial parthenogenesis**, the unfertilized egg (ovum) is induced to develop into a complete individual by physical or chemical stimuli. e.g., Annelid and seaurchin eggs.

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